Causality and Machine Learning:
Probabilistic Graphical Models

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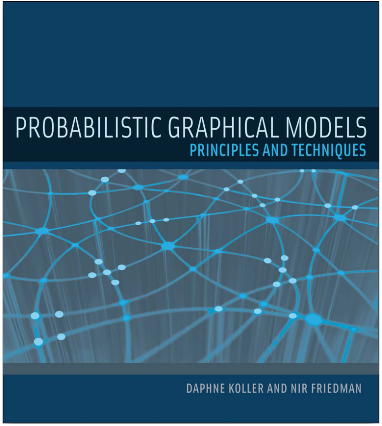
Motivation

- **Goal:** Model a joint distribution P over a set of random variables $X_1, \ldots X_n$.
- **Challenge:** Even in the simple case of binary variables, there are 2^n joint configurations(!), for which an explicit representation is unmanageable, both computationally and statistically.
- **PGM solution:** Make use of the independence properties in the distribution, and represent the (in)dependence structure using a graph.



Overview

1. Probability Theory

Outcome Space, Events, and Event Space

- The set of possible outcomes of an experiment is called the outcome space Ω and an event α is any subset of the outcome space, that is, $\alpha \subseteq \Omega$.
- **Example:** When throwing a six-sided die, we have that $\Omega = \{1, 2, 3, 4, 5, 6\}$ and an event could be $\alpha_{\text{odd}} = \{1, 3, 5\}$.
- **Example:** For a given course, we can define $\Omega = \{\text{"all students enrolled in the course"}\}$ and $\alpha_{\text{A}} = \{\text{"all students that got grade A"}\}$.
- For a given Ω, we collect all the events to which we are willing to assign probabilities into an event space $\mathcal{S}$, which is assumed to be a sigma algebra[1].

[1] It contains the empty set $\varnothing$ and is closed under union and complementation.

Probability Distribution

- **Definition:** A probability distribution P over $(\Omega, \mathcal{S})$ is a mapping from events in $\mathcal{S}$ to real values that satisfies the following conditions (the Kolmogorov axioms):
  1. $P(\alpha) \geq 0$ for all $\alpha \in \mathcal{S}$,
  2. $P(\Omega) = 1$,
  3. If $\alpha, \beta \in \mathcal{S}$ and $\alpha \cap \beta = \varnothing$, then $P(\alpha \cup \beta) = P(\alpha) + P(\beta)$.

- The above conditions imply many other useful conditions; in particular, we have that

$$P(\varnothing) = 0 \quad \text{and} \quad P(\alpha \cup \beta) = P(\alpha) + P(\beta) - P(\alpha \cap \beta).$$

Random Variable

- A random variable X is a function that maps each outcome $\omega \in \Omega$ to a value in the outcome space of the variable, $\texttt{Dom}(X)$, which is typically a set of real numbers.[2]

- The probability function for X is induced by the probability function on Ω:

$$P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\}).$$

- **Example:** In the die throwing scenario we define a binary variable $X : \{\{1, 3, 5\}, \{2, 4, 6\}\} \to \{\text{odd}, \text{even}\}$.

- **Example:** For the student population we define a grade variable G that maps each student to his/her grade, where e.g. $\texttt{Dom}(G) = \{A, B, C\}$.

[2] We will focus on categorical variables, for which $\texttt{Dom}(X)$ is often defined as a set of labels.

Joint and Marginal Distribution

- The joint distribution over a set of random variables, denoted by $P(X_1, \ldots, X_n)$, assigns a probability to each joint configuration $(x_1, \ldots, x_n)$.
- Let $(\mathbf{Y}, \mathbf{Z})$ be a partition of $\{X_1, \ldots, X_n\}$. The marginal distribution over $\mathbf{Y}$ is given by

$$P(\mathbf{Y} = \mathbf{y}) = \sum_{\mathbf{z} \in \texttt{Dom}(\mathbf{Z})} P(\mathbf{Y} = \mathbf{y}, \mathbf{Z} = \mathbf{z}).$$

		Intelligence *low*	*high*	
Grade	*A*	0.07	0.18	0.25
	B	0.28	0.09	0.37
	C	0.35	0.03	0.38
		0.7	0.3	1

Figure: Example of a joint distribution and implied marginal distributions.

Conditional Distribution

- The conditional distribution of X given $Y = y$ is defined by

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}, \quad \text{provided that } P(Y = y) > 0.$$

- **Example:**

$$\begin{gathered} P(Intelligence = high) = 0.3 \\ \neq \\ P(Intelligence = high \mid Grade = A) = \frac{0.18}{0.25} = 0.72 \end{gathered}$$

- **Note:** We use $P(x \mid y)$ as shorthand notation for $P(X = x \mid Y = y)$ and $P(X \mid Y)$ for the set of conditional distributions for X given Y, i.e, $P(X \mid Y = y)$ for all $y \in \mathtt{Dom}(Y)$.

Chain Rule

- From the definition of conditional probability, we directly obtain the chain rule:

$$P(x, y) = P(x)P(y \mid x),$$

  or more generally:

$$P(x_1, \dots, x_n) = P(x_1)P(x_2 \mid x_1) \cdots P(x_n \mid x_1, \dots, x_{n-1}).$$

- We often express this on the distribution level:

$$P(X_1, \dots, X_n) = P(X_1)P(X_2 \mid X_1) \cdots P(X_n \mid X_1, \dots, X_{n-1}).$$

Bayes' Rule

- From the definition of conditional probability, we directly obtain Bayes' rule:

$$P(x \mid y) = \frac{P(y \mid x)P(x)}{P(y)}, \quad \text{provided that } P(y) > 0.$$

- Or more generally, for sets of variables $\mathbf{X}, \mathbf{Y}$:

$$P(\mathbf{x} \mid \mathbf{y}) = \frac{P(\mathbf{y} \mid \mathbf{x})P(\mathbf{x})}{P(\mathbf{y})}, \quad \text{provided that } P(\mathbf{y}) > 0.$$

Independence of Random Variables

- **Definition:** Let $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ be disjoint sets of variables. We say that $\mathbf{X}$ is conditionally independent of $\mathbf{Y}$ given $\mathbf{Z}$ in a distribution P if

$$P(\mathbf{x} \mid \mathbf{y}, \mathbf{z}) = P(\mathbf{x} \mid \mathbf{z}) \quad \text{or} \quad P(\mathbf{y}, \mathbf{z}) = 0$$

  for all $\mathbf{x} \in \mathtt{Dom}(\mathbf{X})$, $\mathbf{y} \in \mathtt{Dom}(\mathbf{Y})$, $\mathbf{z} \in \mathtt{Dom}(\mathbf{Z})$. This is denoted by $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$.

- **Note:** If $\mathbf{Z} = \emptyset$, then we say that $\mathbf{X}$ and $\mathbf{Y}$ are marginally independent.

- **Proposition:** The distribution P satisfies $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$ if and only if

$$P(\mathbf{X}, \mathbf{Y} \mid \mathbf{Z}) = P(\mathbf{X} \mid \mathbf{Z}) P(\mathbf{Y} \mid \mathbf{Z}).$$

Properties of Conditional Independence

- Symmetry: $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) \Rightarrow (\mathbf{Y} \perp \mathbf{X} \mid \mathbf{Z})$.
- Decomposition: $(\mathbf{X} \perp \mathbf{Y}, \mathbf{W} \mid \mathbf{Z}) \Rightarrow (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$.
- Weak Union: $(\mathbf{X} \perp \mathbf{Y}, \mathbf{W} \mid \mathbf{Z}) \Rightarrow (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}, \mathbf{W})$.
- Contraction: $(\mathbf{X} \perp \mathbf{W} \mid \mathbf{Z}, \mathbf{Y})$ & $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) \Rightarrow (\mathbf{X} \perp \mathbf{Y}, \mathbf{W} \mid \mathbf{Z})$.
- For a positive distribution[3] we also have the following property:

  Intersection: $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}, \mathbf{W})$ & $(\mathbf{X} \perp \mathbf{W} \mid \mathbf{Z}, \mathbf{Y}) \Rightarrow (\mathbf{X} \perp \mathbf{Y}, \mathbf{W} \mid \mathbf{Z})$.

[3] A distribution P is positive if all joint configurations have a positive probability.

Example: Proof of the Decomposition Property

- **Task:** Show that $(\mathbf{X} \perp \mathbf{Y}, \mathbf{W} \mid \mathbf{Z}) \Rightarrow (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$ holds.
- **Solution:** We have that for every configuration $(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w})$:

$$
\begin{aligned}
P(\mathbf{x}, \mathbf{y} \mid \mathbf{z}) &= \sum_{\mathbf{w}} P(\mathbf{x}, \mathbf{y}, \mathbf{w} \mid \mathbf{z}) \\
&= \sum_{\mathbf{w}} P(\mathbf{x} \mid \mathbf{z}) P(\mathbf{y}, \mathbf{w} \mid \mathbf{z}) \\
&= P(\mathbf{x} \mid \mathbf{z}) \sum_{\mathbf{w}} P(\mathbf{y}, \mathbf{w} \mid \mathbf{z}) \\
&= P(\mathbf{x} \mid \mathbf{z}) P(\mathbf{y} \mid \mathbf{z})
\end{aligned}
$$

2. Graphs

Nodes, Edges, and Paths

- A graph $\mathcal{G} = \langle V, E \rangle$ is a structure made up of a set of nodes, V, and a set of edges, E.
- In a graphical model, the nodes represent variables, and following earlier notation we generally denote these by $X_1, \ldots, X_n$.
- A pair of nodes X_i, X_j can be connected by a directed edge $X_i \rightarrow X_j$ or an undirected edge $X_i - X_j$ (or $X_j - X_i$), and E contains all such edges.
- We assume that there can only be one edge between a pair of variables, and we use $X_i \rightleftharpoons X_j$ to represent the case where there is an edge but the type is not specified.
- We say that a sequence of nodes, $X_1, \ldots, X_k$, form a path in the graph $\mathcal{G}$ if, for every $i = 1, \ldots, k-1$, we have that $X_i \rightleftharpoons X_{i+1}$.[4]

[4] Referred to as a trail in the PGM book.

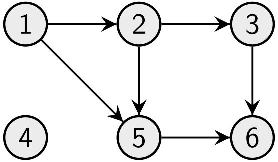
Directed Acyclic Graphs

- We will focus on directed acyclic graphs (DAGs), in which all edges are directed and there are no directed cycles (i.e., no directed paths[5] that starts and ends at the same node).
- Some commonly used structural features in a given DAG $\mathcal{G} = \langle V, E \rangle$:
  - Parents of node X_i: $\mathtt{Pa}(X_i) = \{X_j : X_j \to X_i \in E\}$,
  - Children of node X_i: $\mathtt{Ch}(X_i) = \{X_j : X_i \to X_j \in E\}$,
  - Ancestors of node X_i: $\mathtt{An}(X_i) = \{X_j : \text{there exists a directed path from } X_j \text{ to } X_i \text{ in } \mathcal{G}\}$,
  - Descendants of node X_i: $\mathtt{De}(X_i) = \{X_j : \text{there exists a directed path from } X_i \text{ to } X_j \text{ in } \mathcal{G}\}$.

[5] All edges in a directed path follow the same direction.

Example: DAG

- For example, in the DAG below we have that:
  - $\mathtt{Pa}(X_5) = \{X_1, X_2\}$,
  - $\mathtt{Ch}(X_2) = \{X_3, X_5\}$,
  - $\mathtt{An}(X_3) = \{X_1, X_2\}$,
  - $\mathtt{De}(X_2) = \{X_3, X_5, X_6\}$,
  - $\mathtt{Pa}(X_4) = \mathtt{Ch}(X_4) = \mathtt{An}(X_4) = \mathtt{De}(X_4) = \varnothing$.



3. Exploiting Independence Properties

Example: Independent Random Variables

- Let X_i represent the experiment of tossing coin i with $\texttt{Dom}(X_i) = \{0, 1\}$.
- In this case it is reasonable to assume that mutual independence among $X_1, \ldots, X_n$, implying the factorization:

$$P(X_1, \ldots, X_n) = P(X_1)P(X_2) \cdots P(X_n).$$

- Letting $\theta_i = P(X_i = 1)$, we can represent the joint distribution using only n parameters:

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} \theta_i^{X_i}(1 - \theta_i)^{1-X_i},$$

while an explicit ("independence-free") representation would require $2^n - 1$ parameters.

Example: The Conditional Parameterization

- Assume we want to model the joint distribution over a student's:
  - Intelligence (I), with the possible outcomes *low* (i^0) and *high* (i^1),
  - SAT score (S), with the possible outcomes *low* (s^0) and *high* (s^1).
- The joint distribution $P(I, S)$ has 4 configurations, and can be specified directly:

I	S	$P(I,S)$
i^0	s^0	0.665
i^0	s^1	0.035
i^1	s^0	0.06
i^1	s^1	0.24.

(1 multinomial distribution, 3 free parameters)

Example: The Conditional Parameterization (cont)

- Alternatively, we obtain a more natural representation of the joint distribution using the chain rule:

$$P(I, S) = P(I)P(S \mid I),$$

  which is more compatible with the causal process (although this is not required).

- Instead of specifying joint probabilities, we now specify the marginal probability distribution $P(I)$ and the conditional probability distribution (CPD) $P(S \mid I)$:

i^0	i^1
0.7	0.3

I	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

(3 binomial distributions, 3 free parameters)

Example: Exploiting Conditional Independence

- In addition to I and S, assume that we also have access to the student's grade (G) in some course with the possible values A, B, C (encoded as g^1, g^2, g^3).
- In this case, the joint distribution $P(I, S, G)$ has $2 \cdot 2 \cdot 3 = 12$ configurations, requiring 11 free parameters to represent.
- **Question:** Can we make use of any independence statements to make the representation more compact?
- It is quite natural to assume that $(I \not\perp S)$, $(I \not\perp G)$, but also that $(S \not\perp G)$.

Example: Exploiting Conditional Independence

- It is quite plausible (or a reasonable approximation) to assume that $(S \perp G \mid I)$.
- Using the above CI statement, we can write the joint distributions as

$$P(I, S, G) = P(I)P(S \mid I)P(G \mid I, S) = P(I)P(S \mid I)P(G \mid I).$$

- To fully specify the joint distribution under the above factorization, we need to specify three CPDs:

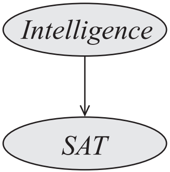
i^0	i^1
0.7	0.3

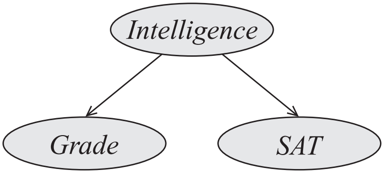
I	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

I	g^1	g^2	g^3
i^0	0.2	0.34	0.46
i^1	0.74	0.17	0.09

(3 binomial and 2 multinomial distributions, 7 free parameters)

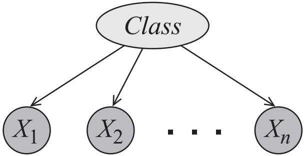
Assumed Graph Structures in the Student Examples





$$P(I, S) = P(I)P(S \mid I) \qquad P(I, S, G) = P(I)P(G \mid I)P(S \mid I)$$

The Naive Bayes Model



- The previous examples are instances of the so-called Naive Bayes model; there is a target class variable C and n feature variables $X_1, \ldots, X_n$.
- The model assumes ("naively") that $(X_i \perp \mathbf{X}_{-i} \mid C)$, where $\mathbf{X}_{-i} = \{X_1, \ldots, X_n\} \backslash X_i$, implying the following factorization:

$$P(C, X_1, \ldots, X_n) = P(C)P(X_1, \ldots, X_n \mid C) = P(C) \prod_{i=1}^{n} P(X_i \mid C).$$

4. Bayesian Networks

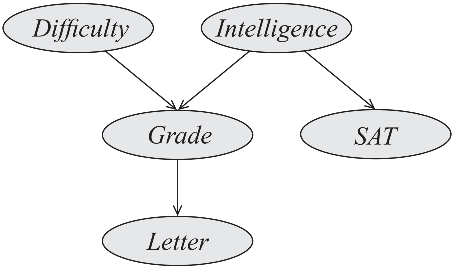
Bayesian Network Structure

- A Bayesian network (BN) builds upon the same intuition as a Naive Bayes model; CI properties of the distribution are exploited to enable a compact representation.
- The core of the Bayesian network representation is a directed acyclic graph (DAG).
- **Definition:** A BN structure is a DAG $\mathcal{G}$ whose nodes represent random variables $X_1, \ldots, X_n$. The structure $\mathcal{G}$ encodes the following set of CI assumptions:

$$(X_i \perp \mathtt{NonDe}(X_i) \mid \mathtt{Pa}(X_i)), \qquad i = 1, \ldots, n,$$

which are known as the local independencies in $\mathcal{G}$ and denoted by $\mathcal{I}_\ell(\mathcal{G})$.

Exercise: List the Local Independencies of the Student Network



Chain Rule for Bayesian Networks

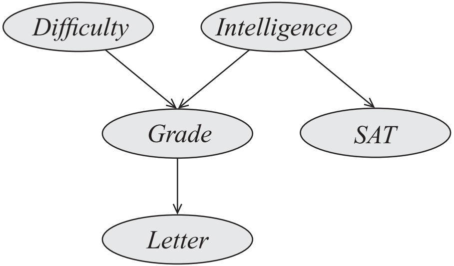
- **Definition:** Let $\mathcal{G}$ be a BN structure (i.e. DAG) over the variables $X_1, \ldots, X_n$. We say that a joint distribution over the same variables factorizes according to $\mathcal{G}$ if P can be expressed as a product

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathtt{Pa}(X_i)),$$

which is called the chain rule for Bayesian networks.

- The individual factors $P(X_i \mid \mathtt{Pa}(X_i))$ are referred to as CPDs or local probabilistic models.

Example: Student Network Factorization

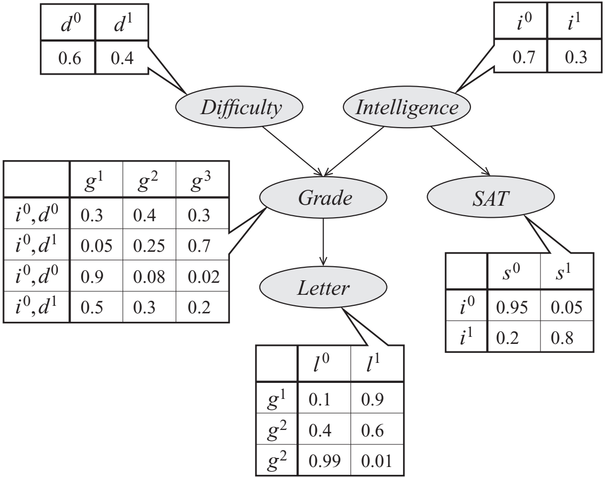


$$P(D, I, G, S, L) = P(D)P(I)P(G \mid D, I)P(S \mid I)P(L \mid G)$$

Bayesian Network

- **Definition:** A Bayesian network (BN) is a pair $\mathcal{B} = \langle \mathcal{G}, P \rangle$ where P factorizes over $\mathcal{G}$, and where P is specified as a set of CPDs associated with the nodes in $\mathcal{G}$.

Example: Student Network



d^0	d^1
0.6	0.4

i^0	i^1
0.7	0.3

	g^1	g^2	g^3
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^0,d^0	0.9	0.08	0.02
i^0,d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^2	0.99	0.01

Two Different Views of the DAG

- A graph $\mathcal{G}$ can be viewed in two quite different ways:

1. A compact representation of a set CI assumptions about a joint distribution P:

$$(X_i \perp \mathtt{NonDe}(X_i) \mid \mathtt{Pa}(X_i)), \qquad i = 1, \dots, n.$$

2. A data structure that encodes how to compactly represent P in a factorized way:

$$P(X_1, \dots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathtt{Pa}(X_i)).$$

- The two views are in a strong sense equivalent.

Independency-Map (I-Map)

- **Definition:** Let P be a distribution over $X_1, \ldots, X_n$ and let $\mathbf{Y}, \mathbf{Z}, \mathbf{W}$ be disjoint subsets of $\{X_1, \ldots, X_n\}$. We define $\mathcal{I}(P)$ to be the set of CI statements of the form $(\mathbf{Y} \perp \mathbf{Z} \mid \mathbf{W})$ that holds in P.
- **Definition:** Let P and $\mathcal{G}$ be a distribution and BN structure, respectively, over $X_1, \ldots, X_n$. We say that $\mathcal{G}$ is an I-map for P if $\mathcal{I}_\ell(\mathcal{G}) \subseteq \mathcal{I}(P)$.

Exercise: Which $\mathcal{G}$:s are I-Maps for $P(X, Y)$?

X	Y	$P(X,Y)$
x^0	y^0	0.08
x^0	y^1	0.32
x^1	y^0	0.12
x^1	y^1	0.48

(a)

X	Y	$P(X,Y)$
x^0	y^0	0.4
x^0	y^1	0.3
x^1	y^0	0.2
x^1	y^1	0.1

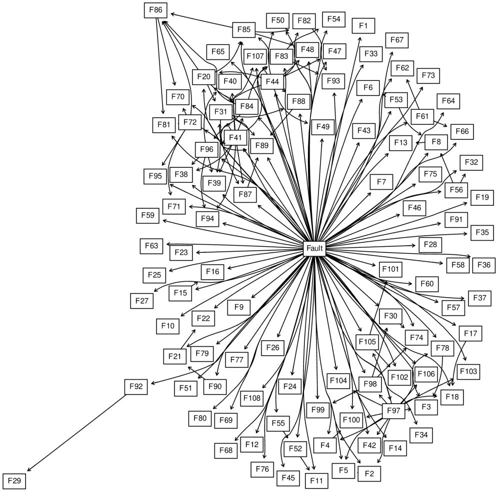
(b)

Factorization and I-Map

- **Theorem:** Let $\mathcal{G}$ be a BN structure over $X_1, \dots, X_n$ and let P be a distribution over the same set of variables. Then, $\mathcal{G}$ is an I-map for P if and only if P factorizes according to $\mathcal{G}$.

Application: The Pathfinder Expert System

- A medical diagnosis system:
  - 1 disease node (60 diseases),
  - 100 features,
  - $\sim$ 70,000 parameters.
- Built up by hand with the help of an expert pathologist.
- Was at least as good as the expert designing the system and better than less experienced pathologists.



Heckerman, Horwitz, and Nathwani (1992). Towards Normative Expert Systems: Part I. The Pathfinder Project. *Methods of Information in Medicine.*

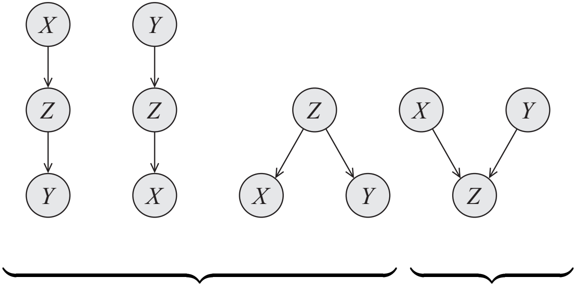
5. d-Separation

Independencies in Graphs

- Understanding the dependencies and independencies of an underlying distribution is important from both a user perspective and an algorithmic perspective.
- We have that a graph $\mathcal{G}$ encodes a set of local CI statements $\mathcal{I}_\ell(\mathcal{G})$.
- Given that P factorizes over $\mathcal{G}$, we have that P satisfies $\mathcal{I}_\ell(\mathcal{G})$ (and vice versa).
- **Question:** Are there any additional non-local independencies implied by $\mathcal{G}$? If so, can such a statement be read off $\mathcal{G}$ directly?

Indirect Connections - Active Paths (Basic Case)

- Consider the simplest case of an indirect connection: $X \rightleftharpoons Z \rightleftharpoons Y$, and think of the paths in a graph as channels through which information can flow.
- When information can flow from X to Y via Z, we say that a path is active.

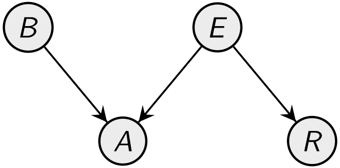


Path is active iff Z is not observed.

Path is active iff Z or a descendant of Z is observed.

Example: Burglar Alarm

- **Scenario:** A burglar alarm (A) in a remote cabin can be set off by a burglary (B) or an earthquake (E), and the owner might hear about the earthquake on the radio (R).



Indirect Connections - Active Paths (General Case)

- Consider the case of a general path $X_1 \rightleftharpoons \ldots \rightleftharpoons X_n$ in $\mathcal{G}$.
- For information to flow from X_1 to X_n, it must flow through every node X_i along the two-edge path $X_{i-1} \rightleftharpoons X_i \rightleftharpoons X_{i+1}$.
- **Definition:** Let $\mathcal{G}$ be a BN structure, and $X_1 \rightleftharpoons \ldots \rightleftharpoons X_n$ a path in $\mathcal{G}$. Let $\mathbf{Z}$ be a subset of observed variables. The path $X_1 \rightleftharpoons \ldots \rightleftharpoons X_n$ is active given $\mathbf{Z}$ if:
  1. Whenever we have a v-structure $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$, then X_i or one of its descendants are in $\mathbf{Z}$,
  2. No other node along the path is in $\mathbf{Z}$.

d-Separation

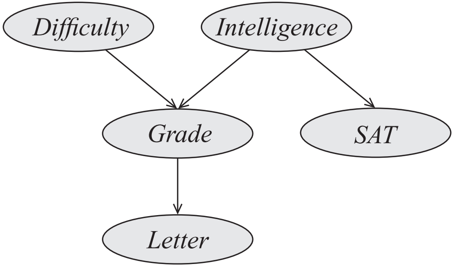
- By considering all possible paths between two nodes X_1 and X_n and checking if they are active, we arrive at the notion of d-separation.

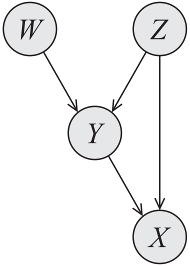
- **Definition:** Let $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ be three sets of nodes in $\mathcal{G}$. We say that $\mathbf{X}$ and $\mathbf{Y}$ are d-separated given $\mathbf{Z}$, denoted by $\text{d-sep}_{\mathcal{G}}(\mathbf{X}\,;\mathbf{Y} \mid \mathbf{Z})$, if there is no active path between any node $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ given $\mathbf{Z}$.

- **Definition:** The global Markov independencies is defined as the set of independencies that correspond to d-separation:

$$\mathcal{I}(\mathcal{G}) = \left\{(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) \colon \text{d-sep}_{\mathcal{G}}(\mathbf{X}\,;\mathbf{Y} \mid \mathbf{Z})\right\}$$

Exercise: Active Paths





- Is there an active path between D and S when:
  (a) no variable is observed?
  (b) L is observed?
  (c) L and I are observed?

- Is there an active path between X and W when:
  (a) no variable is observed?
  (b) Y is observed?

Soundness of d-Separation

- **Theorem:** If a distribution P factorizes over $\mathcal{G}$, then $\mathcal{I}(\mathcal{G}) \subseteq \mathcal{I}(P)$.

- In other words, any independence reported by d-separation w.r.t. $\mathcal{G}$ must hold in P.

- **Question:** What about the other direction? Will every independence that holds in P imply d-separation in $\mathcal{G}$? (A notion known as completeness)

Exercise: Independencies in Distributions

- **Task:** Let X, Y, Z be three binary variables and let their dependence structure be encoded by the following DAG: $Z \rightarrow X \rightarrow Y$. Give an example of a distribution $P(X, Y, Z)$ that factorizes over the DAG and where: (i) $(X \not\perp Y)$ and (ii) $(X \perp Y)$.

Faithfulness

- To formalize completeness for d-separation, we need the notion of faithfulness.
- **Definition:** A distribution P is faithful to $\mathcal{G}$ if, whenever $(X \perp Y \mid \mathbf{Z}) \in \mathcal{I}(P)$, then $\text{d-sep}_{\mathcal{G}}(X\,;Y \mid \mathbf{Z})$.
- Thus, if we have a distribution P that factorizes over $\mathcal{G}$ and is faithful to $\mathcal{G}$, then we have that $\mathcal{I}(P) = \mathcal{I}(\mathcal{G})$ (and we say that $\mathcal{G}$ is a perfect map for P).
- **Remember:** "P factorizing over $\mathcal{G}$" $\not\Rightarrow$ "P faithful to $\mathcal{G}$".

Completeness of d-Separation

- **Theorem:** Almost all distributions P that factorize over $\mathcal{G}$ are also faithful to $\mathcal{G}$, that is, we have that $\mathcal{I}(P) = \mathcal{I}(\mathcal{G})$.[6]

- Thus, d-separation is in a "strong" sense complete.

[6] Specifically, all distributions except for a set of measure zero in the space of the most common CPD parameterizations

An Algorithm for d-Separation

- A naive way of verifying if X and Y are d-separated by $\mathbf{Z}$ is to enumerate all paths and check if they are active.
- A much more efficient algorithm (with complexity $O(|V|+|E|)$) works in two phases:
  1. The graph is traversed bottom up marking all nodes that are in $\mathbf{Z}$ or have descendants in $\mathbf{Z}$ (opening up v-structures);
  2. A breadth-first search from X to Y is made, stopping the traversal along a path if a blocked node is reached.[7,8] If Y is reached, there is an active path between X and Y.

[7] A breadth-first search starts at X and traverses the graph by exploring all of the neighbour nodes at the present depth before moving on to the nodes at the next depth level.

[8] A node is blocked if it is an unmarked middle node in a v-structure or if it is a node in a non-v-structure that is in $\mathbf{Z}$.

6. Markov Equivalence

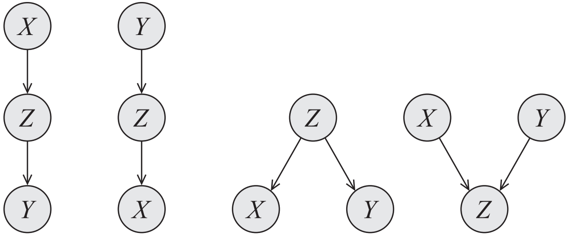
Markov Equivalence

- Viewing a graph, $\mathcal{G}$, as a compact encoding of a set of independencies, $\mathcal{I}(\mathcal{G})$, means that different graphs can be equivalent in terms of the independencies they encode.

- **Definition:** Two graph structures $\mathcal{G}_1$ and $\mathcal{G}_2$ over $X_1, \ldots, X_n$ are Markov equivalent if $\mathcal{I}(\mathcal{G}_1) = \mathcal{I}(\mathcal{G}_2)$. The set of all graphs over $X_1, \ldots, X_n$ is partitioned into a set of mutually exclusive and exhaustive Markov equivalence classes.

- Markov equivalence of two graphs, $\mathcal{G}_1$ and $\mathcal{G}_2$, implies that any distribution P that can be factorized over $\mathcal{G}_1$ can also be factorized over $\mathcal{G}_2$ (and vice versa).

Example: Markov Equivalence



$\mathcal{I}(\mathcal{G}) = \{(X \perp Y \mid Z)\}$ $\qquad$ $\mathcal{I}(\mathcal{G}) = \{(X \perp Y)\}$

A Sufficient and Necessary Condition for Markov Equivalence

- **Definition:** The skeleton of a DAG $\mathcal{G}$ is the undirected version of $\mathcal{G}$.

- **Definition:** A v-structure $X \rightarrow Z \leftarrow Y$ is an immorality if there is no direct edge between X and Y.

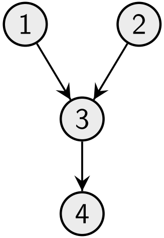
- **Theorem:** Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be two graphs over $X_1, \ldots, X_n$. Then $\mathcal{G}_1$ and $\mathcal{G}_2$ are Markov equivalent if and only if they have the same skeleton and the same set of immoralities.

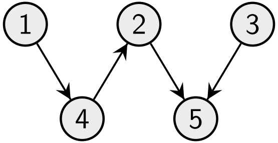
Representing I-Equivalence Classes

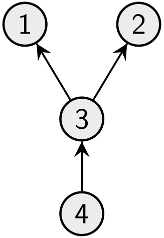
- To compactly represent a Markov equivalence class we use a partially directed acyclic graph (PDAG).
- **Definition:** Let $\mathcal{G}$ be a DAG. A PDAG $\mathcal{K}$ is a completed PDAG (CPDAG)[9] of the Markov equivalence class of $\mathcal{G}$ if it has the same skeleton as $\mathcal{G}$, and contains a directed edge $X \rightarrow Y$ if and only if all $\mathcal{G}'$ that are Markov equivalent to $\mathcal{G}$ contain the edge $X \rightarrow Y$.
- A CPDAG represents potential edge orientations; if an edge is directed, then all members in the equivalence class agree on the orientation.
- **Note:** An edge that is not part of an immorality might also be oriented in a CPDAG.

[9] Referred to as a class PDAG in the PGM book.

Exercise: Markov Equivalence and CPDAGs







7. Summary

Summary

- Bayesian networks: a probabilistic model for the joint distribution, $P(X_1, \ldots, X_n)$.
  - Uses a DAG, $\mathcal{G}$, to encode the dependence structure, $\mathcal{I}(\mathcal{G})$, over the variables;
  - Factorizes the joint distribution according to the DAG;
  - The factorization and the dependence structure are tightly connected.
- d-Separation: a graphical criterion for reading of non-local CIs directly from $\mathcal{G}$.
  - Is based on the notion of active paths;
  - Is sound and complete (under faithfulness);
  - Can be checked with an algorithm that is linear in the number of nodes and edges.
- Markov Equivalence: different DAGs that encode the same dependence structure.
  - Can be checked using the concepts of skeleton and immoralities;
  - Has implications for causal discovery (day 4).